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Incremental Learning in Network Traffic Management: A Fixed-Representation Rehearsal Approach

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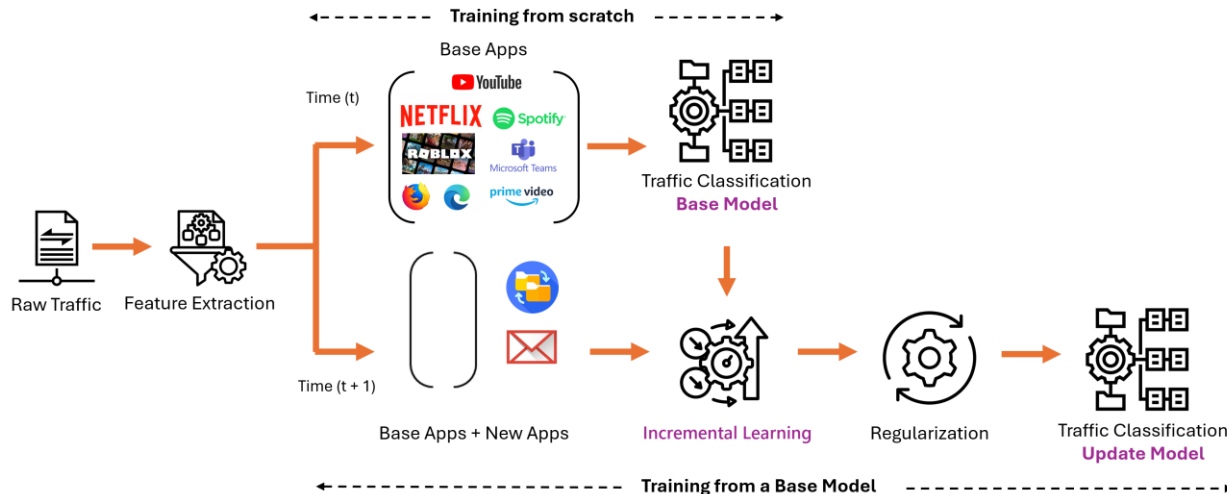
Fixed-Representation Rehearsal — Approach & Results

The Problem

- Network traffic management systems must continuously **evolve to new applications without degrading the performance** of previously learned categories.
- But traffic never sits still: **new apps keep appearing** — the model must be **updated**.
 - **Retrain from scratch**: best accuracy, but costly and requires all historical data.
 - **Fine-tune on new apps only**: low cost but **forgets** the old ones.

Our Approach

- Freeze the **FCNN backbone**, retrain only the **classification head**, replay a **10% buffer**.
- A bounded-cost update — no full retrain, no forgetting. Stages below ↓



Results — from scratch vs incremental

Training time

Time to train the model

from scratch → incremental : reduction

7 → 6 + 1 classes : **1,696 s** → **133 s** : **92.16%**

8 → 6 + 2 classes : **1,872 s** → **192 s** : **89.72%**

Storage

Memory of the training data

from scratch → incremental : reduction

7 → 6 + 1 classes : **9.2 MB** → **2.1 MB** : **76.88%**

8 → 6 + 2 classes : **10.6 MB** → **3.4 MB** : **67.50%**

Forgetting

Accuracy lost on old classes when adding new

old-class accuracy, from scratch → incremental : gap

7 → 6 + 1 classes : **86.4%** → **85.8%** : **0.56%**

8 → 6 + 2 classes : **86.0%** → **84.9%** : **1.13%**

Find Out More!

Come and see how we update traffic classifiers without full retraining!!

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QR code to poster

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